

Department of Urban and Regional Planning
Khulna University of Engineering & Technology (KUET)
URP 3182: GIS and Remote Sensing Studio

Project-02

You are given the following spatial datasets with BTM Projection for Rajshahi City Corporation (RCC)

- a. Education Facilities
- b. Ward Boundaries

1. Prepare RCC boundary (polygon) using spatial data of Ward Boundaries.
2. Taking 500m buffer as standard service area of primary school,
 - (a) Calculate percentage of only served area, overserved area and unserved area within the RCC area.

Status	Area_km2	Percentage
Overserved Area	26.340348	54.84
		23.06
Served Area	11.07640719	
Unserved Area	10.62329346	22.11
Grand Total	48.04004865	100.00%

(b) Calculate percentage of served area, overserved area and unserved area within the RCC area against each ward with reference to total RCC area

Ward No	Overserved Area	Served Area	Unserved Area	Grand Total
2	4.54%	3.76%	6.25%	14.55%
3	1.92%	2.68%	3.33%	7.93%
4	2.21%	0.42%	0.24%	2.87%
5	2.72%	0.26%	0.35%	3.33%
6	2.57%	0.79%	1.18%	4.54%
7	3.57%	1.17%	0.28%	5.01%
8	2.74%	5.47%	3.86%	12.08%
9	2.33%	3.85%	2.91%	9.08%
10	2.30%	0.27%	0.00%	2.56%
11	2.08%	0.06%	0.00%	2.14%
12	1.57%	0.00%	0.00%	1.57%
13	1.62%	0.00%	0.00%	1.62%
14	1.35%	0.02%	0.00%	1.37%
15	1.40%	0.09%	0.01%	1.50%
16	0.98%	0.00%	0.00%	0.98%
17	1.69%	0.57%	2.01%	4.28%
18	1.68%	1.70%	0.50%	3.87%
19	3.14%	0.95%	1.13%	5.21%
20	0.87%	0.00%	0.00%	0.87%
21	0.85%	0.00%	0.00%	0.85%
22	0.68%	0.38%	0.06%	1.12%
23	0.57%	0.00%	0.00%	0.57%
24	0.95%	0.00%	0.00%	0.95%
25	0.62%	0.36%	0.01%	0.99%
26	0.70%	0.00%	0.00%	0.70%
27	0.67%	0.01%	0.00%	0.67%
28	1.01%	0.00%	0.00%	1.01%
29	1.30%	0.00%	0.00%	1.30%
30	3.32%	0.19%	0.01%	3.51%
31	2.90%	0.07%	0.00%	2.98%
Grand Total	54.83%	23.06%	22.11%	100.00%

(c) Calculate percentage of served area, overserved area and unserved area within the RCC area against each ward with reference to Ward area.

Ward No	Overserved Area	Served Area	Unserved Area	Grand Total
2	31.23%	25.83%	42.94%	100.00%
3	24.25%	33.75%	42.00%	100.00%
4	76.97%	14.75%	8.28%	100.00%
5	81.67%	7.84%	10.49%	100.00%
6	56.61%	17.39%	26.00%	100.00%
7	71.16%	23.27%	5.57%	100.00%
8	22.71%	45.31%	31.98%	100.00%
9	25.60%	42.38%	32.02%	100.00%
10	89.48%	10.52%	0.00%	100.00%
11	97.32%	2.68%	0.00%	100.00%
12	100.00%	0.00%	0.00%	100.00%
13	100.00%	0.00%	0.00%	100.00%
14	98.85%	1.15%	0.00%	100.00%
15	93.40%	6.12%	0.48%	100.00%
16	100.00%	0.00%	0.00%	100.00%
17	39.64%	13.29%	47.07%	100.00%
18	43.25%	43.87%	12.87%	100.00%
19	60.19%	18.18%	21.63%	100.00%
20	100.00%	0.00%	0.00%	100.00%
21	100.00%	0.00%	0.00%	100.00%
22	61.18%	33.72%	5.10%	100.00%
23	100.00%	0.00%	0.00%	100.00%
24	100.00%	0.00%	0.00%	100.00%
25	62.49%	36.70%	0.82%	100.00%
26	100.00%	0.00%	0.00%	100.00%
27	98.79%	1.21%	0.00%	100.00%
28	100.00%	0.00%	0.00%	100.00%
29	100.00%	0.00%	0.00%	100.00%
30	94.49%	5.30%	0.21%	100.00%
31	97.44%	2.50%	0.06%	100.00%
Grand Total	54.83%	23.06%	22.11%	100.00%

(d) Calculate the percentage of population for underserved area, overserved area and unserved area within the RCC area.

Status	Overserved	Served	Unserved_	Total
Total Percentage	76.12%	13.11%	10.77%	100.00%

(e) Calculate the percentage of population for underserved area, overserved area and unserved area against each ward with reference to total RCC area

Word No	Overserved	Served	Unserved	Grand Total
1	3.41%	0.40%	0.00%	3.81%
2	0.73%	1.02%	1.26%	3.01%
3	3.79%	0.10%	0.00%	3.89%
4	1.12%	0.38%	1.33%	2.83%
5	3.84%	0.00%	0.00%	3.84%
6	4.84%	0.06%	0.00%	4.90%
7	1.52%	1.54%	0.45%	3.51%
8	1.71%	0.94%	0.14%	2.80%
9	0.91%	0.54%	0.01%	1.46%
10	1.76%	0.12%	0.01%	1.88%
11	2.16%	0.00%	0.00%	2.16%
12	2.91%	0.00%	0.00%	2.91%
13	1.79%	0.00%	0.00%	1.79%
14	1.20%	0.39%	0.09%	1.69%
15	2.52%	0.00%	0.00%	2.52%
16	4.88%	0.47%	0.63%	5.98%
17	1.20%	0.99%	1.65%	3.85%
18	2.88%	0.55%	0.31%	3.75%
19	2.16%	0.66%	0.99%	3.82%
20	4.88%	0.00%	0.00%	4.88%
21	1.04%	0.00%	0.00%	1.04%
22	2.79%	0.00%	0.00%	2.79%
23	2.41%	0.00%	0.00%	2.41%
24	2.50%	0.03%	0.00%	2.53%
25	3.08%	0.00%	0.00%	3.08%
26	0.90%	1.49%	1.12%	3.51%
27	3.99%	1.21%	1.43%	6.63%
28	5.03%	0.28%	0.01%	5.32%
29	3.21%	0.08%	0.00%	3.30%
30	0.93%	1.86%	1.31%	4.10%
Grand Total	76.12%	13.11%	10.77%	100.00%

(f) Calculate the percentage of population for underserved area, overserved area and unserved area against each ward with reference to Ward area.

Row Labels	Overserved	Served	Unserved	Grand Total
1	89.48%	10.52%	0.00%	100.00%
2	24.25%	33.75%	42.00%	100.00%
3	97.32%	2.68%	0.00%	100.00%
4	39.64%	13.29%	47.07%	100.00%
5	100.00%	0.00%	0.00%	100.00%
6	98.85%	1.15%	0.00%	100.00%
7	43.25%	43.87%	12.87%	100.00%
8	61.18%	33.72%	5.10%	100.00%
9	62.49%	36.70%	0.82%	100.00%
10	93.40%	6.12%	0.48%	100.00%
11	100.00%	0.00%	0.00%	100.00%
12	100.00%	0.00%	0.00%	100.00%
13	100.00%	0.00%	0.00%	100.00%
14	71.16%	23.27%	5.57%	100.00%
15	100.00%	0.00%	0.00%	100.00%
16	81.67%	7.84%	10.49%	100.00%
17	31.23%	25.83%	42.94%	100.00%
18	76.97%	14.75%	8.28%	100.00%
19	56.61%	17.39%	26.00%	100.00%
20	100.00%	0.00%	0.00%	100.00%
21	100.00%	0.00%	0.00%	100.00%
22	100.00%	0.00%	0.00%	100.00%
23	100.00%	0.00%	0.00%	100.00%
24	98.79%	1.21%	0.00%	100.00%
25	100.00%	0.00%	0.00%	100.00%
26	25.60%	42.38%	32.02%	100.00%
27	60.19%	18.18%	21.63%	100.00%
28	94.49%	5.30%	0.21%	100.00%
29	97.44%	2.50%	0.06%	100.00%
30	22.71%	45.31%	31.98%	100.00%
Grand Total	76.12%	13.11%	10.77%	100.00%

(g) Calculate the percentage of population within overserved area based on the availability of services like double served, triple served and the like.

(h)

Ward No	2	3	4	5	6	7	8	9	10	11	12	13	14	15	23	Grand Total
1	16	36	56	45	6		16		10	22						207
2	20	24	28	5												77
3	8	42	104	90	66	42	32	18	20	22						444
4	16	36	44	20	18			9								143
5	4	24	68	75	90	91	32	18	10	22				15		449
6	12	36	44	45	42	28	16						14			237
7	18	33	40	35	36	14	16	9	10	11						222
8	8	3	4				8									23
9	12	18	8													38
10	18	27	36	30	30	21		9		11						182
11	6	33	40	10					10							99
12	6	24	36	60	60	42	40	36	10			13	14	15		356
13	2	27	68	65	36	7	8	9		22						244
14	14	69	76	90	96	70	40	45	10	11		13			23	557
15	4	39	100	140	120	63	40	18	10	22	12		14			582
16	34	60	56	50	60	42	8	18		22		13				363
17	34	48	32	15	6	7										142
18	20	30	40	25	12			9	10			13				159
19	22	27	32	35	18				10							144
20		3	24	65	138	119	56	36	20	22	12		14			509
21	2	6	24	70	90	70	24	18		11						315
22		3	24	55	84	133	88	90	20	33		13	14			557
23	2	24	36	40	30	35	24	9					14			214
24	6	18	40	30	6			9	10							119
25	2	30	44	75	48	14			10							223
26	14	9														23
27	26	66	80	85	36			9		11	12					325
28	26	63	104	120	102	35		9			12					471
29	16	33	84	100	108	119	80	27		11	24	13	14		23	652
30	18	24	12					9								63
Grand Total	386	915	1384	1475	1338	952	528	414	170	253	72	78	98	30	46	8139